

The instance booting process in OpenStack is a complex orchestration involving multiple core services that coordinate to transform a user request into a running virtual machine (VM). 1, 2

The Boot Workflow Steps

The process follows a sequential path to ensure resources are allocated correctly and securely:

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1. **Request Submission & Authentication:** A user initiates a request via the **Nova API** or **Horizon Dashboard**. 3-5 The **Keystone service** must first authenticate the user and provide a token to authorize the resource request. 6-8
2. **Scheduling:** The request is sent to the **Nova-scheduler**, which identifies the optimal physical host (compute node) for the VM based on resource availability. 9-11
3. **Database Management:** The **Nova-conductor** manages the necessary database updates related to the new instance without allowing the compute nodes direct access. 12, 13
4. **Resource Allocation:** The system allocates all required "building blocks," including the VM image from **Glance**, network connectivity from **Neutron**, and storage volumes from **Cinder**. 1, 3
5. **Execution:** The **nova-compute** service on the selected host receives the instruction via the message bus and performs the final task of starting the virtual machine. 11

The Image Retrieval Process

Images in OpenStack are treated as **templates** for VM deployments. 9, 14

- **Glance Service:** Acts as the central repository where images are stored and retrieved. 15, 16
- **Retrieval:** During the boot process, Nova requests the specific image (in formats like QCOW2, RAW, or VMDK) from Glance. 1, 15
- **Customization:** Users can inject custom scripts or configuration data during this stage using tools like **Cloud-Init** or **User Data** to configure the VM as it starts. 17, 18

The Role of the Nova-Scheduler

The scheduler acts as the "brain" of the placement process. 10 It uses a three-step **Filter Scheduler** logic: 19

1. **Filtering:** It eliminates unsuitable compute nodes based on requirements like CPU, RAM, disk space, and specialized hardware (e.g., GPUs). 20, 21
2. **Weighting:** It ranks the remaining suitable nodes. Nodes with more free resources or those strategically placed for better network performance are given higher weights. 21-23
3. **Selection:** The node with the highest weight is selected as the destination for the VM. 23

The Role of the Compute Node (Nova-Compute)

The compute node is the physical server that provides the actual processing power and memory. 24

- **Service Daemon:** The nova-compute daemon runs on the node, listening for instructions on the **message bus**. 11
- **Hypervisor Interaction:** It interacts with the hypervisor (like KVM, VMware, or Hyper-V) to access the hardware layer and launch the VM. 25
- **Security Isolation:** For security, the compute node is **denied direct access to the database**, relying instead on the nova-conductor for data interactions. 11, 25

Network Configuration During Boot

The **Neutron** service serves as the "nerve center" for connectivity during the boot process: 26

- **Provisioning:** It creates virtual switches and subnets to establish a communication path. 26, 27
- **IP Management:** It assigns IP addresses to the VM through **DHCP** or static allocation. 28, 29
- **Port Creation:** Each VM is assigned **Ports**, which serve as the attachment points to the virtual network. 30
- **Security Groups:** Neutron applies virtual firewalls (security groups) to the VM's ports to control incoming and outgoing traffic from the moment it starts. 28, 30, 31

Exam-Ready Explanation (10-Mark Answer Guide)

- **Introduction:** Define booting as the multi-service process of starting a VM, requiring authentication, host selection, and resource allocation. 1, 17
- **The Workflow (Sequential):**
- **User/API:** Request entry and Keystone authentication. 6, 7
- **Scheduler:** The decision engine using Filtering and Weighting. 21, 23
- **Image (Glance):** Template retrieval and configuration injection (Cloud-Init). 15, 18
- **Network (Neutron):** Provisioning switches, assigning IPs, and applying security groups. 27, 28
- **Compute Node:** The nova-compute process executing the VM launch on the physical hardware. 11
- **Key Component Roles:**
- **Nova-Conductor:** Acts as a secure database proxy. 13, 32
- **Nova-Scheduler:** Optimizes resource allocation to avoid bottlenecks. 23, 33
- **Nova-Compute:** Manages the VMM/Hypervisor lifecycle. 11, 25
- **Conclusion:** Effective booting ensures standardized deployments, high availability (via Availability Zones), and a secure cloud posture. 34, 35